Samarth Sharma

Major: Electrical Engineering
Minors: Industrial & Management Engineering, English Literature

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023-Present	Bachelors of Technology	Indian Institute of Technology Kanpur	**8.6**/10
2023	CBSE(XII)	Hope Hall Foundation School (HHFS), Delhi	**96.8**%
2021	CBSE(X)	Delhi Public School, Prayagraj	**97.4**%

Scholastic Achievements

- **Reviewer**, **SPCOM** 2026 (International Conference on Signal Processing and Communications), Indian Institute of Science.
- Secured an **All India Rank** of **1082** in the **Joint Entrance Examination (Advanced)**, 2023 among 200,000 candidates.
- Secured an **All India Rank** of **954** in the **Joint Entrance Examination (Main)**, 2023 among 1.2 million candidates.
- Awarded '**Outstanding Achievement in Academics**' for placing second overall in school in CBSE XII Boards. (2022-23)

Internship Experience

Integration and Automation of Optimized Flop Analysis Flow using Implementation Design Check (IDC)
Hardware Engineering Intern, RTL Design, Chip Innovation and Infrastructure (CI2), **Google India** *(May'26-Present)*

Objective	• To design, analyze, and deploy an **RTL-end LLM-assisted** flop waiver analysis workflow for designers.
Approach	• Integrated Synopsys' **Implementation Design Check (IDC)** tool to Google's internal design and PD workflow for high-impact **TPU networking** projects **left-shifting** flop analysis saving significant crunch time. • Designed a **Gemini-powered interactive multi-agent** framework for **automated waiver generation**, leveraging IDC's built-in register-RTL mapping reports to limit manual RTL source tracing while enabling **natural-language** waiver specification and designer-guided validation of grouped optimization patterns. • **Benchmarked** IDC against existing optimization-analysis methodologies, evaluating register violation correlation, designer productivity, and waiver coverage with respect to **Fusion Compiler** (FC). • Collaborated with **international RTL, Physical Design** teams to identify and report flow gaps to Synopsys.
Impact	• Deploying an intelligent RTL-end automated waiver generation and **root-cause analysis framework**, reducing turnaround time from over **300 hours to under 8 hours** by limiting RTL source tracing to unresolved registers. • Served as a **primary technical point of contact** conducting tutorials, demos, one-on-one debugging sessions with **RTL and physical designers** across teams, enabling adoption before major **design milestones**. • Authored detailed **support documents**, guides, and **changelists** (CLs) for efficient knowledge transfer.

Deep Learning-Aided Radar Signal Processing: Algorithmic Augmentation and FPGA Acceleration
Research Intern, Prof. Sumit J. Darak, Algorithms to Architecture (A2A) lab, **IIIT Delhi** *(May'25-Apr'26)*

Objective	• To design low-latency, low-complexity, high-performance **deep-learning-aided** radar signal processing (RSP) algorithms and architectures on **Zynq-MPSoCs** for Integrated Sensing and Communication (**ISAC**) applications.
Approach	• Developed low-latency custom pipelined **HLS IPs** and integrated **PYNQ** drivers using **AXI4** interfaces. • **Augmented** Doppler and DoA algorithms using **MLPs** and **CNNs** after detailed **ablation studies**. • Integrated **Quantization-aware Fine Tuning** (QAFT) and **Word Length Optimization** to the augmented architecture to reduce BRAM and DSP utilization offering superiority both in terms of latency and resources. • Designed a **novel** light-weight **end-to-end (E2E)** MLP architecture for reliable Doppler estimation. • Created a new **OOR** (out-of-range) metric to better capitulate robustness of the proposed architecture. • Used **Integrated Logic Analyzers** to debug AXI transactions and verify dataflow correctness on hardware.
Impact	• Co-authored **two international conference** papers and **one journal** manuscript in ISAC and RSP domains. • Attained a **30× acceleration** over NumPy for reliable Doppler estimation using MUSIC and ESPRIT. • Achieved **59.8%** lower RMSE, and **20.4×** lower latency using novel E2E MLP based architectures over classical algorithms, while using just **20 slow-time packets** to match the **170** packet performance at lower SNRs. • Designed a novel weight-reconfigurable architecture, providing an **additional 56% performance gain**. • Created a **state-of-the-art** QAFT architecture using **INT8** weights to reduce BRAM utilization by **81.9**%. • Published a **4-part YouTube tutorial series** on design flow and benchmarking on **Vitis 2024.2**.

Research Publications

- *Low Latency Deep Learning Doppler Velocity Estimation for Integrated Sensing and Communication* (First Author), **International Conference on Signal Processing and Communications (SPCOM) 2026**, IISc. Bangalore (July 2026).
- *Reconfigurable Low-Complexity Architecture for Doppler Velocity Estimation of Tightly-Spaced Mobile Users in ISAC* (Second Author), **IEEE Radar Conference 2026** in Phoenix, Arizona, USA (May 2026). arXiv
- *Deep Learning Augmented Doppler and Direction of Arrival Estimation for Integrated Sensing and Communication* (Co-first author) Targeting **IEEE Transactions journals**, manuscript in preparation

Key Projects

RTL design of a Mixed-Signal Backend Controller | Prof. Chithra | Course Project, EE619 | IIT Kanpur *(Jan'26-Apr'26)*

- Designed a Verilog **backend controller** for a mixed signal IC with FSM-based startup sequencing and serial communication.
- Implemented **CDC**-safe multi-clock design and verified functionality using a comprehensive testbench and **GTKWave**.

- Performed synthesis and timing analysis, evaluating timing constraints using open-source tools like **Yosys** and **OpenSTA**.

Transistor-Level Duty Cycle Correction Circuit on Cadence Virtuoso | Prof. Chithra | IIT Kanpur *(May'25-Jun'25)*

- Designed a **transistor-level** duty cycle correction circuit in **gpdk-180nm** CMOS technology on **Cadence Virtuoso**.
- Implemented a **bidirectional shift register (BSR)** and **XOR-delay-DFF** based feedback loop for adaptive convergence.
- Achieved a corrected duty cycle of **48–52%** for **0.5–1 GHz** inputs, starting from initial distortions ranging between **30–70%**.

Semiconductor Device Modeling using DEVSIM TCAD | Prof. Rituraj | IIT Kanpur (*Dec'24-Feb'25*)

- Explored **DEVSIM TCAD** framework, including device setup, **meshing**, and model definition using Python scripting.
- Simulated 1D **p-n junction diode** using **drift-diffusion** equations; analyzed **IV behavior** and **carrier** dynamics.
- Explored solver mechanics, **convergence** behavior via **Newton's method**, and visualized results using **Matplotlib**.

Technical Skills

Programming Languages: Verilog HDL, C, C++, Python, LaTeX; **Software:** Xilinx Vitis HLS, Vivado, Cadence Virtuoso, Yosys, OpenSTA, MATLAB, MicroCap, GNU Octave, Devsim TCAD, Altium

Relevant Coursework

*: A/*A**(10/10) #: PG level elective *o*: Online Course

Electronics	**Computer Science**	**Mathematics**	**Electrical and Physics**
VLSI System Design#	Data Structures and Algorithms	Probability and Statistics	Communication Systems*
Digital Electronics	Fundamentals of Computing	Complex Variables*	Signals, Systems and Networks
Microelectronics-II*	Harvard's CS50x*o	Partial Differential Eqns	Quantum Physics
Microprocessors, Digital, Electronic Circuits Lab	Machine Learning*o	Differential Equations*	Classical Electrodynamics
Spin-Electronics Devices#*	Natural Language Processing*o	Linear Algebra	Control Systems

Positions of Responsibility (PoRs) and Volunteering

- **Academic Department Mentor, EE** - UG Academics Wing, Academics and Career Council, IITK (2025-26 tenure): Coached ∼**200** EE sophomores guiding them through course work, projects, internship season, and resume building.
- **Academic Mentor**- Centre for Mental Health and Well-being, IITK (2024-25 tenure): Mentored ∼**500** first-year students in **Quantum Physics** (PHY114) by conducting remedial sessions and providing personalized academic guidance.
- **Secretary**, Debating Society (DebSoc), IIT Kanpur (2024-25 tenure): Assisted in organizing flagship **national-level** debating tournaments like *IITK APD'24* (online) and *IITK BPD'25* (offline); **led marketing** efforts for *IITK APD'24*.
- **Volunteer**- National Service Scheme (NSS), IIT Kanpur (2023-24). Contributed in the field of **education** for socio-economically underprivileged youth by translating physics lectures into regional languages to increase accessibility.